

SIM1002 Calculus I
Tutorial 1
Functions and Their Graphs

Exercise 1.1

1. Find the domain and range of each function.
 - (a) $f(x) = 1 - \sqrt{x}$
 - (b) $g(t) = \sqrt{x^2 - 3x}$
 - (c) $G(t) = \frac{2}{t^2 - 16}$
2. Find the domain of $y = \frac{x+3}{4 - \sqrt{x^2 - 9}}$.
3. Find the range of $y = 2 + \sqrt{9 + x^2}$.
4. Graph the following equations and explain why they are not graphs of functions of x .
 - (a) $|y| = x$,
 - (b) $|x| + |y| = 1$,
 - (c) $|x + y| = 1$.
5. Graph the functions.
 - (a) $F(x) = \begin{cases} 4 - x^2, & x \leq 1 \\ x^2 + 2x, & x > 1. \end{cases}$
 - (b) $G(x) = \begin{cases} \frac{1}{x}, & x < 0 \\ x, & x \geq 0. \end{cases}$
6. What real numbers x satisfy the equation $\lfloor x \rfloor = \lceil x \rceil$?
7. Graph the function
$$f(x) = \begin{cases} \lfloor x \rfloor, & x \geq 0 \\ \lceil x \rceil, & x < 0. \end{cases}$$
Why is $f(x)$ is called the *integer part* of x ?
8. In the following determine whether the function is even, odd, or neither.
 - (a) $g(x) = \frac{x}{x^2 - 1}$
 - (b) $h(t) = 2|t| + 1$
 - (c) $g(x) = 1 + \cos x$.

Exercise 1.2

1. Find domains of $f, g, \frac{f}{g}, \frac{g}{f}$, where $f(x) = 1, g(x) = 1 + \sqrt{x}$.

2. If $f(x) = x - 1$ and $g(x) = \frac{1}{x+1}$, find the following.

(a) $f\left(g\left(\frac{1}{2}\right)\right)$ (b) $g\left(f\left(\frac{1}{2}\right)\right)$

(c) $f(g(x))$ (d) $g(f(x))$

(e) $f(f(2))$ (f) $g(g(2))$

(g) $f(f(x))$ (h) $g(g(x))$

3. Copy and complete the following table.

	$g(x)$	$f(x)$	$(f \circ g)(x)$
a.	$\frac{1}{x-1}$	$ x $	?
b.	?	$\frac{x-1}{x}$	$\frac{x}{x-1}$
c.	?	$\sqrt{x}$	$ x $
d.	$\sqrt{x}$	?	$ x $

4. Give an equation for the shifted graph. Then sketch the original and shifted graphs together. Label each graph with its equation.

(a) $x^2 + y^2 = 25$, Up 3, left 4.

(b) $y = x^{\frac{2}{3}}$, Right 1, down 1,

5. Graph the functions in the following.

(a) $y = \sqrt{9-x}$

(b) $y = (x-8)^{\frac{2}{3}}$

(c) $y = \frac{1}{x+2}$

(d) $y = \frac{1}{(x+1)^2}$

6. Give an equation for the stretches or compressed graph.
- (a) $y = x^2 - 1$, compressed horizontally by a factor of 2.
 - (b) $y = 1 + \frac{1}{x^2}$, stretched horizontally by a factor of 3.
 - (c) $y = \sqrt{4 - x^2}$, compressed vertically by a factor of 3.
 - (d) $y = 1 - x^3$, stretched horizontally by a factor of 2.

Exercises 1.3

1. Express the given quantity in terms of $\sin x$ and $\cos x$.
- (a) $\sin(2\pi - x)$
 - (b) $\cos\left(\frac{3\pi}{2} + x\right)$
2. Evaluate $\cos\left(\frac{11\pi}{12}\right)$ as $\cos\left(\frac{\pi}{4} + \frac{2\pi}{3}\right)$.
3. Evaluate $\sin\left(\frac{5\pi}{12}\right)$.
4. Solve for the angle θ , where $0 \leq \theta \leq 2\pi$.
- (a) $\sin^2 \theta = \cos^2 \theta$
 - (b) $\sin 2\theta - \cos \theta = 0$.